

Can Brain Signals Tell Learning Models Apart?

Decoding Reward Prediction Error from EEG with Three Reinforcement-Learning Models, Two Decoders and Two Datasets

Final Research Report

Written by a second-year undergraduate student

Abstract

When people learn from rewards, several different mathematical models can explain their choices almost equally well. I wanted to know if the brain's electrical activity (EEG) can help decide which model is closer to the truth. I fitted three learning models to the choices of 23 people: Rescorla-Wagner (M1), a dual-learning-rate model (M2) and a Pearce-Hall model (M3). From each model I got a trial-by-trial reward prediction error (RPE, meaning how surprising the outcome was). Then I trained small neural networks to predict that RPE from feedback-locked EEG in three scalp regions, always testing on a person the network had never seen.

The main finding on the first dataset (ds004295) is a small but robust effect in the parieto-occipital region: an interaction between the predicted RPE and reward that is present for M1 and M3 but not for M2. It held in 4 training seeds, survived leaving out each person in turn (23 of 23), survived a global correction over 90 tests, and reappeared in a Transformer once that Transformer was trained properly. But there are important complications. The plain decoding was mostly explained by win-versus-loss, not by graded RPE. M1 and M3 could not be told apart from each other. The behaviour-only comparison changed depending on how it was done: in-sample AIC preferred M2, while out-of-sample prediction found no clear winner. And when I ran a second dataset (ds003458, a three-armed bandit) through a related decoding pipeline, the interaction was **not significant** ($p = 0.17$ for both M1 and M3), so the effect has not been replicated.

So the honest summary is: there is a real but modest signal in one dataset, no confirmed replication, and no proof yet that EEG can tell these learning models apart. The most reliable results of the project are methodological: the reward confound, the in-sample versus out-of-sample disagreement (which appeared in both datasets), and the need to train a decoder properly before calling it a failure.

1. Introduction

1.1 The basic idea

Humans and animals learn by trial and error. A very popular idea in neuroscience is that the brain keeps a running guess of how good each option is, and updates that guess using a **reward prediction error (RPE)**: the difference between what happened and what was expected. If you expected little and got a big reward, the RPE is large and positive. If you expected a reward and got nothing, the RPE is negative. This idea is closely linked to the dopamine system.

1.2 The problem

Many different learning models can be written down, and several of them fit human choices almost equally well. But they can predict different trial-by-trial RPE values. So if we only look at behaviour,

we may not be able to say which model is closer to what the brain does. This is sometimes called **model mimicry**.

1.3 My question

Can competing reinforcement-learning models be told apart by decoding their predicted RPE from EEG, and does any such difference hold across decoder types and across independent datasets?

Related work already exists. Other researchers have used single-trial EEG regression to separate overlapping prediction-error signals (for example Hoy, Steiner and Knight, 2021), and the feedback-related negativity in the frontal-central EEG has long been linked to reward prediction errors. I am not claiming to invent the approach. I apply it to several competing models, with modern decoders and with many checks against fooling myself.

1.4 Words used in this report

Word	Simple meaning
Reinforcement learning (RL) model	A small formula for how a person updates their beliefs about which option is good after each outcome.
RPE (reward prediction error)	Outcome minus expectation. The 'surprise' signal that drives learning.
Learning rate (alpha)	How strongly one outcome changes the belief. High alpha means quick, jumpy learning.
Inverse temperature (beta)	How strictly a person chooses the option they believe is better. Low beta means more random choices.
Rescorla-Wagner (M1)	The simplest model: one learning rate for everything.
Dual learning rate (M2)	Two learning rates, one for wins and one for losses.
Pearce-Hall (M3)	The learning rate changes over time depending on recent surprise.
AIC	A score for how well a model fits, with a penalty for having more parameters. Lower is better.
Out-of-sample	Testing a model on data it was not fitted on.
Model mimicry	Different models fitting the data about equally well.
Parameter recovery	Simulating fake people with known settings and checking if the fitting code finds the same settings back.
EEG epoch	A short piece of EEG cut around an event, here from 0.2 s before to 0.8 s after the feedback.
Leave-one-subject-out (LOSO)	Train on all people except one, test on that one, repeat for everyone.
Decoder	A neural network that predicts something (here RPE) from EEG.
Confound	Something else that changes together with what I care about and can fake the effect. Here, win vs loss.
Partial correlation	A correlation after removing the effect of a third variable (here win/loss).
Interaction	The relationship between two things changes depending on a third. Here: the link between predicted and actual RPE differs between wins and losses.

Word	Simple meaning
FDR correction	A way to avoid false alarms when running many tests at once.
Jackknife	Removing one person at a time and redoing the test to check that no single person drives the result.
Seed	A number that fixes the randomness of training so a run can be repeated.

2. Aim, Questions and Hypotheses

Aim: to test whether different reward-learning models, which explain choices about equally well, can be separated by how cleanly their trial-by-trial RPE can be decoded from EEG, and whether this depends on the scalp region.

2.1 Sub-questions

1. Do the three models fit behaviour about equally well (model mimicry)?
2. For each model, how well can its RPE be decoded from EEG in the frontal, central and parieto-occipital regions?
3. Does combining regions help?
4. Does the effect carry over from reward trials to punishment trials, and back?
5. Do people whose RPE is easier to decode also learn differently?
6. Does anything found in one dataset hold in a second, independent dataset?

2.2 Hypotheses written before the analysis

- **H1:** at least two models fit behaviour so similarly that they cannot be separated (mimicry).
- **H2:** frontal and fronto-central EEG shows the strongest RPE decoding (as in the feedback-related-negativity literature).
- **H3:** no single model wins in every region.
- **H4 (exploratory):** combining regions modestly improves decoding.

I wrote these down first so that any hypothesis that failed could be reported as a failure. A scorecard is in Section 8.7.

3. What I Claim and What I Do Not Claim

I claim	I do NOT claim
On ds004295 there is a robust interaction (predicted RPE x reward) in the parieto-occipital region for M1 and M3, not for M2. It holds across 4 seeds, all 23 leave-one-person-out checks and a global 90-test FDR correction.	That EEG decodes graded RPE. After removing win/loss, the partial correlation is about zero in every region.
The effect also appears in a Transformer once the Transformer is trained properly (early stopping).	That M1 and M3 are neurally different. A direct paired test found no reliable difference between them.
The in-sample AIC winner (M2) does not survive out-of-sample testing. In-sample and out-of-sample results disagree in both datasets I checked.	That the in-sample winner M2 is the best model. Its advantage is probably flexibility.

I claim	I do NOT claim
I built and checked complete behaviour and EEG pipelines on a second dataset, and ran the decoding there.	That the finding replicates. On ds003458 the interaction was not significant ($p = 0.17$), and the two pipelines are not identical.
Everything here is reported with its complications: what failed, what disagrees, and what is still open.	That I know which model the brain uses.

4. Data

4.1 Dataset 1 (main): ds004295

From Stolz, Pickering and Mueller. People did two probabilistic reversal-learning tasks. In the **reward** task, positive feedback meant +10 cents and negative feedback meant no money. In the **punishment** task, positive feedback meant avoiding a loud white-noise burst and negative feedback meant the burst was played (its loudness was set per person to match the reward). In both tasks, one of two options was correct most of the time and the correct option switched now and then, so people had to keep learning.

- 26 people are listed. 3 were excluded by the dataset authors (two for muscle artefacts, one for technical EEG problems), so **23 people** remain.
- Each person has **560 trials** (280 reward + 280 punishment), so **12,880 trials** in total.
- EEG: BioSemi ActiveTwo, 1024 Hz, 66 channels in the raw files (64 scalp + 2 eye channels). I cut epochs from 0.2 s before to 0.8 s after feedback (1025 time points), with baseline correction.
- I used three regions: frontal (8 channels), central (8), parieto-occipital (9), so 25 channels in total.

4.2 Dataset 2 (replication attempt): ds003458

From the Cavanagh lab (University of New Mexico). 23 healthy college students (ages 18 to 24) did a **three-armed bandit** task. The reward chance of each option slowly moved between about 20% and 90%, each with a different phase, so there was no fixed 'correct' option and no reward/punishment split. About 479 trials per person remained after removing trials without a response. EEG had 64 channels, and I made three regions from them: frontal (16), central (20), parieto-occipital (17), with 11 channels (temporal sites and eye channels) left out.

4.3 The two datasets side by side

	ds004295 (main)	ds003458 (replication attempt)
People used	23 (of 26)	23
Task	Two-option reversal learning, reward and punishment blocks	Three-option bandit with slowly changing probabilities
Trials per person	560 (280 + 280)	About 479
Win rate	About 60%	About 75% to 87%
EEG	1024 Hz, 66 channels in raw files	64 channels, epochs at 500 Hz
Regions	Frontal 8, central 8, parieto-occipital 9	Frontal 16, central 20, parieto-occipital 17

	ds004295 (main)	ds003458 (replication attempt)
Epoch	-0.2 to 0.8 s around feedback	-0.2 to 0.8 s around feedback
Decoder testing	Leave-one-person-out, several channels as input	Within-person 5-fold, one averaged trace per region

Table 1. The two datasets. The last row matters: the two decoding designs are not the same (see Section 9).

5. Getting the Behavioural Data Right

This step took much longer than I expected, and it is the reason I trust the rest. Everything in this project depends on knowing exactly which option each person chose on each trial and what feedback they got. If that is wrong, the RL models learn nonsense and the EEG comparison means nothing.

Step	What I did / saw	What it taught me
1	I read choices from the event triggers as 'left' or 'right' and fitted a Rescorla-Wagner model to the first 10 people.	Every fit was at chance level (the model did no better than a coin flip). The fitting code was fine, since simulated people with known settings were recovered correctly (for example true alpha 0.3 came back as 0.29). So the data were wrong.
2	I tried many ways of decoding the choice from the response and selection triggers (position, response code, combinations) and compared their win-stay and lose-shift rates with the values published in the dataset.	None matched the published win-stay values of about 0.79 to 0.83.
3	I plugged the dataset authors' own published model parameters into my likelihood function.	Their parameters also did worse than chance on my choices (for example 256.7 vs a chance value of 194.1 for one block). So my choice sequences were wrong, not my model.
4	I found the authors' code folder with their behavioural files (280 actions and 280 feedback values per block, for 23 people).	This gave ground-truth behaviour. I switched to it.
5	I had to match each of the 23 behaviour records to the right person. Matching by the win-stay statistic and by reaction-time correlation was weak (reaction-time correlation only about 0.26, far below what a real match needs).	Those two checks alone could not confirm the pairing, so I did not rely on them.
6	I matched each person's reward sequence in the behaviour file against the feedback markers in their EEG.	All 23 people matched exactly (100%). This is the check I trust, so the pairing is verified rather than assumed from file order.

Table 2. How the behavioural data were fixed.

Two more small things: one person (sub-s5) has no official win-stay values, so that person is skipped cleanly in the comparison instead of being silently dropped, and every later step checks that trial counts and RPE values match between stored files (the stored M1 RPE and the recomputed M1 RPE differed by exactly 0).

6. The Three Learning Models

Model	How it updates	Free parameters
M1: Rescorla-Wagner	Value of the chosen option moves towards the outcome by a fixed fraction (alpha) of the RPE.	alpha, beta
M2: Dual learning rate	Same, but the fraction is alpha_gain after a win and alpha_loss after a non-win.	alpha_gain, alpha_loss, beta
M3: Pearce-Hall	The learning rate goes up when recent outcomes were surprising and goes down when they were predictable.	alpha0, kappa, eta, beta

Table 3. The three models. Choice follows a softmax rule (for three options in dataset 2).

I fitted each model separately to each person and each block (46 person-blocks in dataset 1), using several random starting points, and got the RPE for every trial from the fitted model. The RPE values of the models are very similar: the average trial-by-trial correlation is **0.970 for M1 vs M3**, 0.938 for M1 vs M2 and 0.906 for M2 vs M3. This will matter later, because two nearly identical signals are hard to separate with EEG.

6.1 Can the fitting code be trusted? (parameter recovery)

I simulated fake people with known settings, fitted the models, and checked whether the fitted values match the true ones (Figure 1).

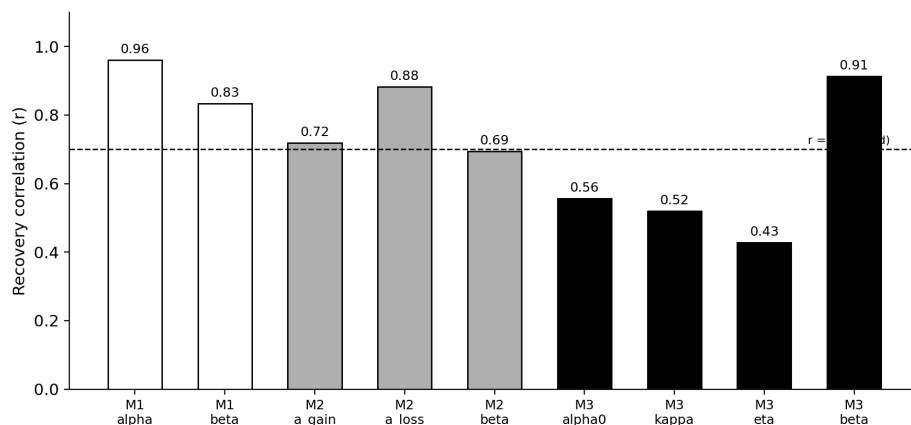


Figure 1. Parameter recovery on dataset 1. Bars above the dashed line ($r = 0.7$) are recovered well.

M1 is recovered well (alpha $r = 0.960$, beta $r = 0.833$). M2 is mixed: alpha_gain 0.719, alpha_loss 0.882, but beta only 0.694. M3 is **weak** for three of its four parameters (alpha0 0.557, kappa 0.520, eta 0.428), and only beta is good (0.913). So I trust M3's RPE output more than I trust the individual fitted values of its parameters. On dataset 2, M1 recovery is excellent (alpha $r = 0.982$, beta $r = 0.985$); I have not yet re-run recovery for M2 and M3 there.

7. EEG Decoding Methods

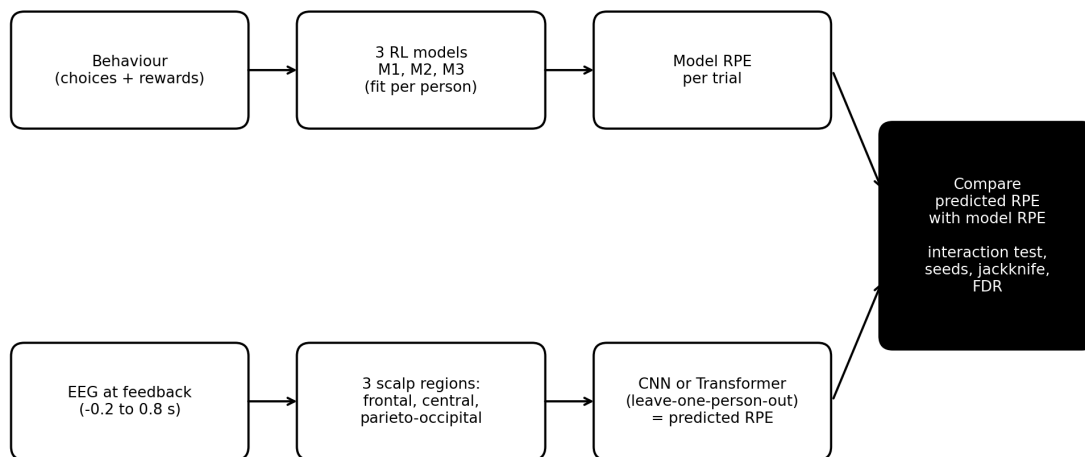


Figure 2. The whole pipeline.

7.1 The decoder

For each region and each model, a small 1D convolutional neural network (CNN) reads the EEG epoch (channels x 1025 time points) and outputs one number, the predicted RPE for that trial. It has two convolution layers (16 and 32 filters), a global average, and one output. It is trained with mean-squared error, Adam (learning rate 0.001), batch size 32 and 30 epochs. Inputs are normalised with the mean and spread of the **training** people only.

Leave-one-subject-out: for each of the 23 people, the CNN is trained on the other 22 and predicts the left-out person's trials. The network never sees the test person. I repeated this with 4 different seeds. Because a network is retrained for every fold, region, model and seed, all predictions were saved to files and reused, so the analysis can be repeated exactly.

7.2 A trap: win versus loss

RPE is strongly tied to whether the trial was a win or a loss. In this data the average absolute correlation between RPE and reward is **0.860**. So a decoder can score a positive correlation with RPE just by learning 'win or loss'. To check this I always report three things:

- **Raw correlation** between predicted and actual RPE.
- **Partial correlation** after removing win/loss (does the decoder carry graded RPE information beyond win/loss?).
- **Interaction test:** I regress actual RPE on predicted RPE, reward, and predicted RPE x reward. The interaction term says whether the link between predicted and actual RPE is different for wins and losses. Each person gets one interaction number, and I test whether the average across the 23 people differs from zero.

Here is an easy way to picture the interaction. Suppose that after a win the decoder's number goes up when the true RPE goes down, but after a loss it goes up when the true RPE goes up. Then the link is 'flipped' between wins and losses, and the interaction is non-zero. It tells me the EEG pattern carries something that depends on outcome type, which is not the same thing as a clean graded RPE signal.

7.3 Other checks I built in

- **Motor confound:** central EEG could reflect the button press, so I checked reaction time.

- **Multiple seeds** for the CNN and the Transformer.
- **Jackknife**: remove each person and redo the test.
- **FDR correction** over one global grid of 90 planned tests in 5 families.
- **A z-scored version of the interaction** plus a permutation test, needed when comparing decoders whose predictions have very different scales (Section 8.4).
- **Baselines**: gradient boosting (HistGradientBoosting and AdaBoost) on 15 hand-made EEG features, to see if the CNN gives anything extra.

8. Results on Dataset 1 (ds004295)

8.1 Behaviour: do the models fit, and which one wins?

People chose the better option and stayed after wins: on average the win rate was 60.4% and the stay rate 80.0%. The average fitted M1 learning rate was 0.41 (very different between people, SD 0.29). The number of person-blocks where a model beat chance was 35 of 46 for M1, 39 of 46 for M2 and 35 of 46 for M3.

In-sample comparison (AIC). M2 won in 27 of 46 person-blocks, M1 in 15 and M3 in 4. Every pair differed by more than the usual mimicry limit (mean absolute AIC difference: M1 vs M2 9.86, M1 vs M3 4.40, M2 vs M3 12.09) and every pair was significant (p between 0.0001 and 0.024). On this view, the models are **not** mimicking each other.

Out-of-sample comparison. Now I fitted each model on one block and tested it on the other (reward to punishment, and the reverse). M2 still won most often by count (13 of 23 people, vs M1 6 and M3 4), but none of the three pairwise differences was significant (M1 vs M2 $p = 0.30$, M1 vs M3 $p = 0.29$, M2 vs M3 $p = 0.83$). A test that ignores outliers gave the same answer (Wilcoxon $p = 0.30$ to 0.36). Four people (sub-s22, s18, s8, s13) had very large differences and had lower fit quality (they beat chance on 62.5% of blocks against 78.95% for the other 19). I report this as a flag, not a reason to remove them, because the group-level result did not change with the robust test.

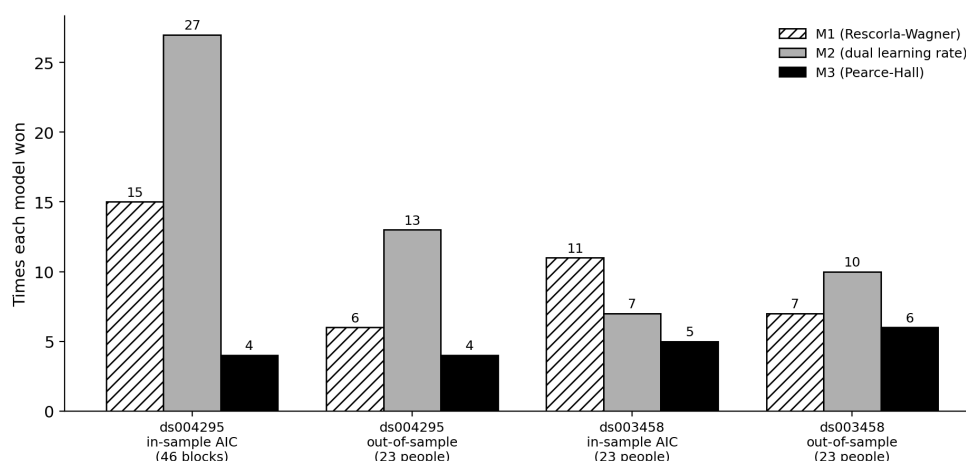


Figure 3. How often each model won, in-sample and out-of-sample, on both datasets. M2's in-sample lead on ds004295 shrinks out-of-sample, and ds003458 shows no clear winner either way.

My reading: M2's in-sample advantage most likely comes from having an extra parameter, not from being truly better. One caution: my out-of-sample test also changes the task (reward to punishment), so it is a hard transfer test. That makes 'no difference' partly a statement about how noisy this kind of test is.

8.2 EEG decoding: mostly win versus loss

First, before any neural network, a simple group-level regression of EEG amplitude on RPE showed a relation in frontal and central regions (after FDR, $p = 0.014$ and 0.024) but not in parieto-occipital ($p = 0.081$). After adding reward to the regression, none survived (all FDR $p = 0.058$). That already hinted at the reward confound.

The CNN shows the same thing (M1, seed 0, Figure 4). The raw correlation between predicted and actual RPE is small but significant in all three regions. It is largest in the frontal region (0.080), then central (0.066), then parieto-occipital (0.047). But the correlation of the predictions with reward alone is even higher (0.101 frontal), and once win/loss is removed the partial correlations are about zero (frontal -0.010, central +0.019, parieto-occipital -0.008; all p above 0.12).

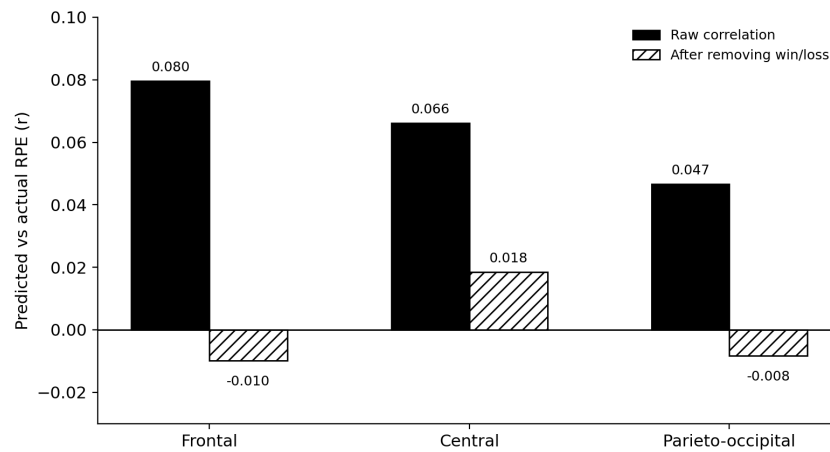


Figure 4. Decoding of M1 RPE. Black bars: raw correlation. Hatched bars: after removing win/loss.

Region (M1, seed 0)	Raw r	Corr. with reward	Partial r	Within wins	Within losses
Frontal	0.0797	0.1006	-0.0098	-0.0187	-0.0038
Central	0.0662	0.0663	0.0185	0.0013	0.0389
Parieto-occipital	0.0467	0.0584	-0.0083	-0.0321	0.0265

Table 4. Average across 23 people. Raw p -values are below 0.003 in all three regions; partial p -values are 0.44 (frontal), 0.13 (central) and 0.49 (parieto-occipital).

Motor check: the central region's raw correlation was 0.0662 and 0.0676 after controlling for reaction time, and the correlation with reaction time itself was near zero (-0.029). So button-press timing does not explain the central result.

Simple baselines: gradient boosting on 15 EEG features gave an even higher raw correlation than the CNN (HistGradientBoosting 0.136, AdaBoost 0.106) but the same pattern after control (partial 0.023 and 0.022, not significant). This shows that the reward confound is a property of the data, not of my CNN.

8.3 The robust parieto-occipital interaction (M1 and M3, not M2)

Although graded RPE was not decodable, the interaction test told a different story in one region. In the parieto-occipital region the interaction was significant for M1 (beta = -0.429, $p = 0.0025$) and M3 (beta = -0.488, $p = 0.0027$), and not for M2 (beta = -0.025, $p = 0.61$). With FDR over the three models, M1 and M3 have $p = 0.0041$ and M2 has $p = 0.61$.

Model	Mean beta across 4 seeds	Seeds with $p < 0.05$	Range of beta
M1	-0.4375	4 of 4	-0.569 to -0.339
M2	-0.0156	0 of 4	-0.026 to +0.002
M3	-0.4373	3 of 4 (seed 3: $p = 0.071$)	-0.533 to -0.284

Table 5. Parieto-occipital interaction across CNN seeds.

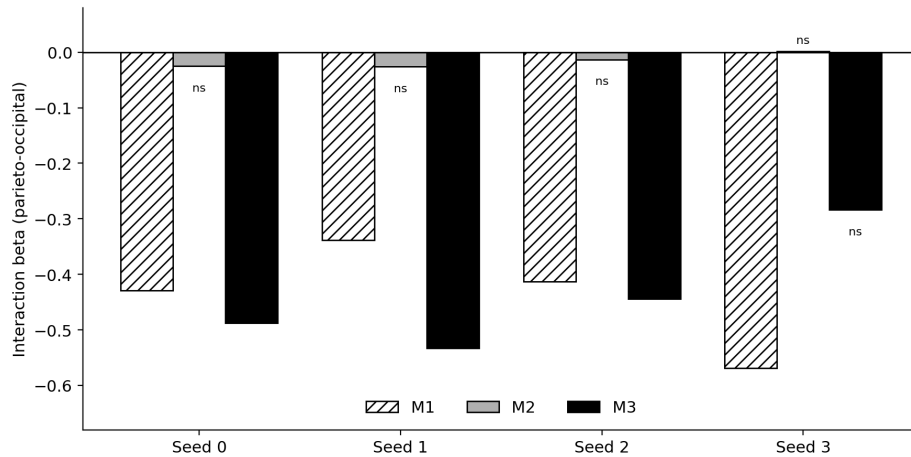


Figure 5. Interaction beta for every seed. 'ns' marks seeds with p above 0.05.

- **Jackknife:** leaving out any one of the 23 people, the effect stayed significant in 23 of 23 runs for both M1 and M3. The most influential person was sub-s20.
- **Global FDR:** across all 90 planned tests, the PO/M1 and PO/M3 interactions survived (adjusted p about 0.013), and so did the small negative PO/M2 partial correlation (-0.047, adjusted $p = 0.012$ to 0.016). In total, 21 of the 90 tests survived.
- **Nature of the effect:** for PO/M1 the correlation within wins is slightly negative (-0.032) and within losses slightly positive (+0.027). Neither is significant alone, but together they make the 'flipped' pattern that the interaction picks up. I treat this as a valence-dependent EEG pattern, not as proof of a graded prediction-error signal.

8.4 Does it depend on the decoder? CNN versus Transformer

I built a compact Transformer (a convolution stem, a learnable position code, 2 attention layers, average over time) with the same leave-one-person-out setup. With 30 fixed training epochs it found **nothing**: the interaction was not significant in any of 4 seeds for M1, M2 or M3.

But its predictions were suspicious. The spread (SD) of the predicted RPE was about 0.35, versus about 0.08 to 0.10 for the CNN and the properly trained version. That is a sign of poor training. I retrained it for up to 80 epochs with a validation person for **early stopping**. After that, the parieto-occipital interaction for M1 appeared in all 4 seeds (z-scored test).

Because the two decoders have prediction values on very different scales, I also used a **z-scored interaction** (predictions standardised per person) and a **permutation test**. On seed 0 the permutation p was 0.0033 for CNN/M1 and 0.0033 for CNN/M3, against 0.80 and 0.36 for the 30-epoch Transformer.

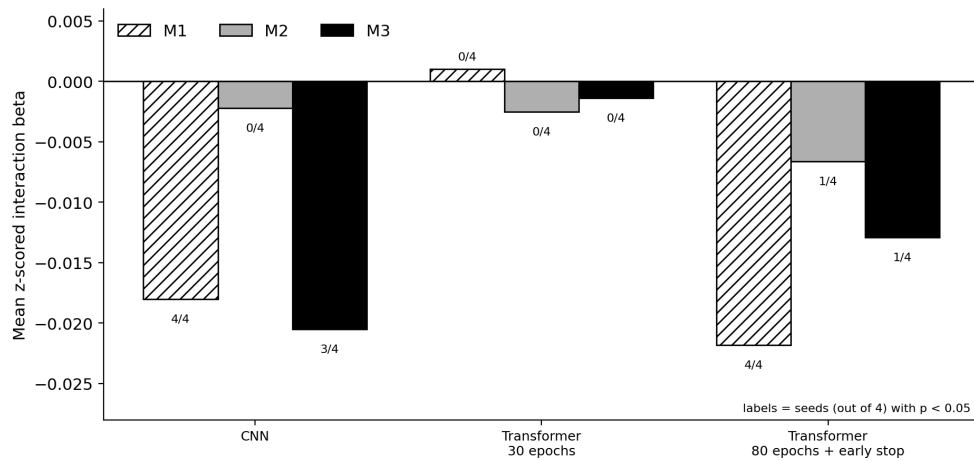


Figure 6. z-scored interaction beta by decoder and model. Labels show how many of 4 seeds had $p < 0.05$.

Two tests, two answers. A raw regression test found M1 significant in only 2 of 4 early-stopped Transformer seeds, while the z-scored, permutation-checked test found 4 of 4. I prefer the z-scored version because it equalises the scale difference, but I show both, because a reader should be able to judge this choice. The lesson I take: a 'negative' result from a decoder that was not trained properly says little about the brain. It says the decoder was not trained properly.

8.5 Are M1 and M3 different from each other?

Being significant for M1 and for M3 separately is not the same as M1 and M3 being different. So I compared the two per-person interaction values directly (paired test):

Comparison	Mean difference	Seeds with $p < 0.05$	Median p
CNN, M1 vs M2	-0.0158	3 of 4	0.018
CNN, M1 vs M3	+0.0025	1 of 4	0.57
Transformer (early stop), M1 vs M2	-0.0152	2 of 4	0.050
Transformer (early stop), M1 vs M3	-0.0089	2 of 4	0.13

Table 6. Paired comparisons of the per-person interaction beta.

M1 and M3 cannot be separated, which fits their 0.970 RPE correlation. M1 differs from M2 clearly under the CNN and more weakly under the Transformer. So the defensible claim is a **two-group** one: single-rate-family models (M1, M3) against the dual-rate model (M2).

8.6 Exploratory analyses (all null or unstable)

Analysis	Result
Combining regions (H4)	Frontal + central + parieto-occipital gave raw $r = 0.092$ vs 0.080 for the best single region, not a significant improvement (paired $p = 0.22$).
Cross-valence (train on reward, test on punishment, and reverse)	Raw correlation for M1 was weakly present in both directions ($p = 0.023$ and 0.004 ; after FDR only punishment-to-reward, $p = 0.025$). After removing win/loss, nothing ($p = 0.24$ to 0.95).
Decodability vs behaviour	24 correlations between decodability and win rate, stay rate, alpha and beta: none survived FDR (smallest raw $p = 0.22$, FDR $p = 0.97$).

Analysis	Result
Clustering people by RL parameters (Gaussian mixture)	BIC chose 3 groups (sizes 20, 2 and 1). The grouping is unstable (adjusted Rand index 0.27, where above 0.7 is stable) and driven by 2 to 3 unusual people, some with beta at the lowest allowed value. So it is not a discovered subgroup structure.

Table 7. Exploratory results.

8.7 Hypothesis scorecard

	Hypothesis	Outcome
H1	At least two models cannot be separated by behaviour	Depends on the test. In-sample: no, all pairs differ. Out-of-sample: yes, no pair differs (ds004295 $p = 0.29$ to 0.83). On ds003458 no pair is significant either way.
H2	Frontal decoding is strongest	Yes for raw decoding (frontal $0.080 > \text{central } 0.066 > \text{parieto-occipital } 0.047$, and the same order on ds003458). But this is win/loss information, since no region carries graded RPE after control.
H3	No single model wins in every region	Partly. Raw decoding favours M1 in all regions. A model difference appears only in the parieto-occipital region (M1, M3 vs M2), and nowhere else.
H4	Combining regions helps a little	No ($p = 0.22$).

Table 8. Scorecard.

9. Replication Attempt on Dataset 2 (ds003458)

9.1 Getting the data right

Dataset 2 did not have a filled trial-by-trial stimulus column, so I rebuilt each trial from the trigger codes: which option was chosen, whether it was a win, and the reaction time. I checked the reconstruction two independent ways (parsing the events files and parsing the raw EEG annotations). Reward agreement was 100%, and chosen-option agreement reached 100% only after I fixed an offset error in my first trigger decoding. I then adapted the three models to a three-option choice rule.

9.2 Behaviour

	In-sample (AIC)	Out-of-sample (odd/even trials)
Winner counts (of 23)	M1 11, M2 7, M3 5	M2 10, M1 7, M3 6
Mean absolute difference or p-values	Mean AIC gaps 5.65 (M1-M2), 4.59 (M1-M3), 7.07 (M2-M3); $p = 0.080, 0.660, 0.208$	$p = 0.83, 0.82, 0.93$
Above chance	M1, M2, M3: 23 of 23	-

Table 9. Behavioural model comparison on ds003458.

The same story appears: no model is a clear winner, and no pair reaches significance in either test. The disagreement between in-sample and out-of-sample results that I found on dataset 1 is not dramatic here (nothing is significant in-sample), but the weak, non-significant separation supports

the view that these models are hard to tell apart by behaviour. I see this cross-dataset result as one of the more robust findings of the project, independent of the EEG result.

9.3 EEG decoding on dataset 2

I high-pass filtered the raw EEG at 0.5 Hz and subtracted a baseline from -0.2 to 0 s. (My very first attempt skipped the filter, and slow drifts dominated the signal, so I redid it.) Each region became one averaged waveform of 500 time points per trial. A small CNN (three convolution blocks) was trained to predict M1 or M3 RPE with **within-person 5-fold cross-validation**. I left M2 out here because it was consistently null on dataset 1. The average decoding correlations were small but positive:

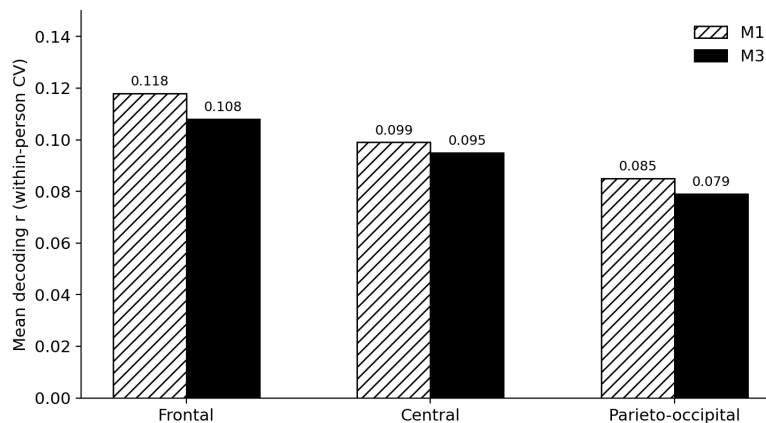


Figure 7. Average decoding correlation per region on ds003458 (23 people). The order frontal > central > parieto-occipital matches dataset 1's raw pattern.

Then I ran the interaction test in the parieto-occipital region, averaging over 4 seeds:

Model	Mean interaction beta	t	p	FDR p
M1	-0.0168	-1.407	0.1735	0.1745
M3	-0.0166	-1.403	0.1745	0.1745

Table 10. Interaction test on ds003458. Jackknife: 0 of 23 leave-one-out runs stay below $p = 0.05$ (largest p 0.314 for M1 and 0.320 for M3).

9.4 What this result means, and what it does not

The sign of the effect is negative, like in dataset 1, but it is **not significant** (p about 0.17), and no leave-one-out run reaches $p < 0.05$. So the parieto-occipital interaction has **not** been replicated. But I cannot say it has been convincingly refuted either, because the two decoding designs differ in ways that matter:

- Dataset 1 used leave-one-person-out decoding with several channels as input. Dataset 2 used within-person shuffled 5-fold decoding with one averaged trace per region. Neighbouring trials are similar, so shuffled folds can leak information and make decoding look better than it is.
- The regression is written in opposite directions (dataset 1: actual RPE on predicted RPE, reward and their product; dataset 2: decoded RPE on true RPE, reward and their product). The betas have different scales (-0.43 vs -0.017), so they cannot be compared as numbers.
- The tasks differ a lot (reversal learning with punishment vs a three-armed drifting bandit with a high win rate), and both datasets have only 23 people, so the tests have limited power.

I therefore call this an **unsuccessful and not-fully-matched replication attempt**. A fair replication would run the dataset-1 pipeline unchanged (multi-channel region tensors, leave-one-person-out, same regression) on dataset 2. That is the top item in Future Work.

10. Mistakes, Traps and Fixes

What went wrong or was risky	How I noticed	How it was handled
Choices coded by screen position	All model fits sat at chance level; simulated data were recovered fine	Switched to the authors' ground-truth behaviour files (Section 5).
Matching behaviour files to people by file order	Win-stay and reaction-time checks were only weakly consistent	Matched by the reward sequence against EEG feedback markers: 23 of 23 exact.
Missing official values for one person	A NaN crashed the comparison	That person is skipped explicitly and listed.
In-sample AIC could overstate the winner	M2 won in-sample but not out-of-sample	Reported both; treat AIC as descriptive.
Reward is almost the same as RPE	Raw decoding worked even where graded RPE did not	Added partial correlation, within-win and within-loss correlations and the interaction test.
Motor activity could fake a central effect	Central EEG sits over the motor area	Controlled for reaction time. No change.
Testing many regions and models	3 regions x 3 models x many metrics	One global FDR grid of 90 tests.
One lucky training run	Deep-learning results change with the seed	4 seeds for the CNN and the Transformer.
A single influential person	Only 23 people	Jackknife over all 23 people.
Transformer trained for too few epochs	Prediction SD 0.35 vs 0.08 to 0.10, and nothing significant	Retrained with early stopping. The effect appeared in 4 of 4 seeds (z-scored).
Two tests disagree on the Transformer	Raw test 2 of 4 seeds; z-scored test 4 of 4	Both reported. Z-scored plus permutation test preferred because it removes a scale confound.
Dataset 2: EEG not filtered at first	Slow drift dominated the signal	0.5 Hz high-pass and baseline subtraction.
Dataset 2: trigger-code offset	The chosen option did not match the events file 100% at first	Corrected the offset in the trigger decoding. Now 100% agreement.
My earlier status report said the dataset 2 EEG decoding was not done	Now it is done	Reported here: no significant interaction ($p = 0.17$).
My earlier notes described the data as '32-channel'	The raw files contain 66 channels	Reported as 66 raw channels, with 25 used in the stored region tensors (8 + 8 + 9). How exactly these 25 were chosen and cleaned is an open item.

What went wrong or was risky	How I noticed	How it was handled
Interaction described as a 'graded signal'	Partial correlations were about zero	Now described as a valence-dependent pattern, not a graded RPE signal.

Table 11. Bug and fix log.

11. What Is Different or Useful About This Project

The idea that RL models might be told apart in the brain is not new. What I think is less common, especially for a student project, is this combination:

- **Behaviour verified against ground truth** instead of assumed from file order, on two structurally different datasets.
- **Parameter recovery before any EEG analysis**, which showed that M3's individual parameters cannot be trusted.
- **A reward-confound-aware analysis** (partial correlations and the interaction test), plus gradient-boosting baselines showing the confound is in the data.
- **A cross-architecture check** (CNN and Transformer), including catching an under-trained Transformer and reporting both significance tests instead of choosing the nicer one.
- **In-sample versus out-of-sample model comparison** treated as two different questions that can disagree.
- **A global 90-test FDR grid, seeds, and a jackknife.**
- **Reporting a failed replication** and its limits, instead of leaving it out.
- **Reproducible pipeline**: every expensive step is cached, so the results can be regenerated.

12. Discussion

What is solid. (1) The three models are very similar in behaviour and in RPE (0.91 to 0.97), so they are hard to separate. (2) Raw EEG decoding of RPE is small and mostly reflects win versus loss. (3) The behaviour-only winner depends on how you test: in-sample AIC favoured M2, out-of-sample did not. (4) Decoder training matters: an under-trained Transformer gave a false 'no effect'.

What is interesting but fragile. The parieto-occipital interaction separates M1 and M3 from M2 in dataset 1, is robust across seeds, people and a second decoder, and survives a big correction. But it is small, it is a valence-dependent pattern rather than a graded RPE signal, and it did not appear in dataset 2. The negative sign in dataset 2 is encouraging, but a fair test is still needed.

A mismatch worth noting. Behaviour (in-sample) prefers M2, while the EEG interaction groups M1 with M3 and puts M2 apart. These two views point in different directions. Since out-of-sample behaviour finds no winner, I do not think this is a contradiction I can resolve yet.

Why might dataset 2 be null? Possible reasons, none tested: differences in the task, a different decoding design, the leak risk of shuffled folds, a lower chance of finding an effect with 23 people, or that the dataset-1 effect is small or specific. Only a same-pipeline rerun can tell these apart.

Overall. This project supports a modest conclusion: EEG carries a small, valence-dependent signal that lines up with the M1/M3 versus M2 split in one dataset, but I do not have a confirmed way of telling the learning models apart from EEG.

13. Real-World Meaning: Why This Could Matter

13.1 Where this kind of work can be useful

Area	How it could help	What is still needed
Basic neuroscience	Choosing between theories of learning when behaviour alone cannot. My work shows a careful way to try this, and shows what can go wrong.	Replication in bigger and more varied datasets.
Computational psychiatry	Learning rate and inverse temperature are studied as markers in depression, anxiety and addiction. Knowing which parameters can be recovered reliably (M1 yes, M3's individual parameters no) tells researchers which numbers to trust.	Larger samples and links with clinical measures.
Brain-computer interfaces and adaptive tutoring	Detecting 'the user just got feedback they did not expect' could let a system adapt its difficulty or feedback.	Much stronger single-trial decoding (today's correlations are about 0.05 to 0.12).
AI and reinforcement learning	If a learning rule is found to match brain data better, it may inspire better agents (the 'residual analysis' idea in my Phase C).	A model that is stable across datasets first.
Research practice	Reward-confound checks, out-of-sample model comparison, seeds, jackknife and FDR are reusable for other EEG studies.	Nothing. They can be used now.

Table 12. Possible uses. These are directions and not things my project has proven.

13.2 What my improvements mean in practice

Improvement	Why it matters outside this project
Verifying behaviour against ground truth	A silent alignment error can create or hide any 'brain effect'. Checking it is cheap compared with a wrong paper.
Partial correlation and interaction tests	Many reported 'RPE signals' in EEG may really be win/loss signals. These tests separate the two.
Out-of-sample model comparison	In-sample fit rewards flexible models. Out-of-sample testing is closer to what we want: a model that predicts new behaviour.
Training the decoder properly	A negative result from a poorly trained network can be mistaken for 'no effect in the brain'.
Global FDR and jackknife	With 23 people and many tests, false positives are easy to produce. These checks lower that risk.
Reporting the failed replication	Others can see exactly how far the finding goes and avoid building on it too early.

13.3 A realistic warning

Nothing here is ready for a clinical or product setting. The effects are small, found in 23 people and not replicated. The realistic value of this project is that it shows how to test this question carefully and where the traps are.

14. Limitations

- **Small samples:** 23 people in each dataset, and RPE from M1 and M3 correlates at 0.97, so both datasets have little power to find subtle differences between models.
- **Not a matched replication:** dataset 2 used a different decoding design (within-person shuffled folds, one trace per region, opposite regression direction), so its null result is not a clean test of the dataset-1 finding.
- **Shuffled folds on dataset 2** can leak information between neighbouring trials; blocked folds would be safer.
- **Graded RPE is not decodable.** The main effect is an interaction, which I interpret as valence-dependent, not graded RPE.
- **M3's parameters are poorly recovered** (three of four are weak) and M2 and M3 recovery on dataset 2 has not been re-checked.
- **The out-of-sample test also changes the task** (reward to punishment), so it mixes model quality with task change.
- **Two significance tests disagree** on the Transformer (raw 2 of 4 vs z-scored 4 of 4). I prefer the z-scored one and say so, but it is a choice.
- **Channel selection:** the raw files have 66 channels, and my stored region tensors use 25. The exact steps behind that selection and the EEG cleaning of the stored tensors are not documented in the notebooks.
- **The final person-matching step** (100% reward-sequence agreement) is not among the notebook cells packaged with this report. The included cells show the earlier, weaker checks, so the notebook should be completed before anyone reruns the pairing.
- **GPU training is not perfectly repeatable** between sessions, so all dataset-1 results come from one consistent set of saved prediction files.
- **Planned but not done:** diffusion-model augmentation of EEG epochs was in my original plan but I did not carry it out.

15. Future Work

15.1 Finish the current study

Idea	Why	How I would check
Run the exact dataset-1 pipeline on dataset 2 (multi-channel tensors, leave-one-person-out, same regression)	This is the fair replication test	Compare the interaction beta and p directly with dataset 1.
Use blocked (contiguous) folds instead of shuffled folds	Stops neighbouring trials leaking	Repeat the dataset-2 decoding and see if raw decoding drops.
Re-check M2 and M3 recovery on dataset 2, and decode M2 there too	Completes the matching of the two datasets	Same recovery test as for M1.
Document the 25-of-66 channel choice and the EEG cleaning	Needed before any final claim	Trace it back to the original preprocessing and write it down.

Idea	Why	How I would check
Pick one primary significance test (z-scored plus permutation) and fix it before new runs	Removes the two-tests problem	Write the rule down first, then run.
Out-of-sample comparison with an odd/even split on dataset 1	Separates model quality from the reward-to-punishment change	Compare with the block-transfer result.

15.2 Make the models and analysis better

Idea	Why	How I would check
Hierarchical (Bayesian) fitting of RL parameters	Better recovery with few trials, especially for M3	Repeat parameter recovery.
Add the authors' other models (for example decay of unchosen options)	M5-type models may fit better	Compare out-of-sample.
Time-resolved analysis (feedback-related negativity and P3 windows)	Finds when the effect appears	Interaction test per time window with FDR.
Use more channels or source estimates	Regions are coarse	Compare against the 3-region result.
Diffusion-based augmentation, only for tests and never for headline numbers	Small trial counts per person	Check against real held-out data.
A third dataset with reversal-learning structure	A second clean replication	Use the same fit-or-reject rules as for the other datasets.

15.3 Towards real-world use

Idea	Why	How I would check
Link fitted parameters to questionnaires and the dataset's noise-rating data	First step towards computational-psychiatry use	Correlate parameters with those measures, with FDR.
Test single-trial 'surprise' detection for a simple adaptive game	A concrete application	Measure hit rate and false alarms in a small live test.
Phase C: analyse where the winning model's predictions differ from the brain data, revise its update rule and test it as an agent against plain Q-learning	Turns the finding into something useful for AI	Only after a stable cross-dataset effect exists.

16. Questions Someone Might Ask

Q: Did you show that EEG can tell the learning models apart?

A: Not yet. In one dataset there is a robust interaction that separates M1 and M3 from M2. It is small, it is not a graded RPE signal, and it was not significant in the second dataset.

Q: Why not just trust AIC?

A: AIC on the same data rewards flexible models. M2 won in-sample, but when tested out-of-sample there was no significant winner in either dataset.

Q: The decoding correlations are around 0.05. Is that meaningful?

A: It is statistically detectable with 12,880 trials, but tiny in size and mostly driven by win versus loss. I do not present it as strong decoding.

Q: Why did the Transformer at 30 epochs show nothing but the later one did?

A: The 30-epoch model was under-trained (predictions with a much larger spread). With early stopping the effect appeared in all 4 seeds under the z-scored test.

Q: Is the dataset 2 result a failed replication?

A: It is an unsuccessful attempt, but not a clean test, because the decoding design was different. A same-pipeline rerun is needed.

Q: Why did you leave M2 out on dataset 2?

A: It was null on dataset 1, so I focused on M1 and M3. This is a choice that a full replication should undo.

Q: Are 23 people enough?

A: For a large effect maybe. For small differences between models, no. Both datasets are under-powered for that.

17. Conclusion

I asked whether competing reinforcement-learning models can be separated by decoding their reward prediction error from EEG, and whether such a separation survives different decoders and a second dataset. On the main dataset I found a small, robust, region-specific interaction that groups M1 with M3 and puts M2 apart, and it survives seeds, a jackknife, a global correction and a properly trained Transformer. I also found that the decoder mostly reads win versus loss, that M1 and M3 cannot be told apart, that behavioural model rankings change between in-sample and out-of-sample tests, and that the interaction did not reach significance on a second dataset.

So my conclusion is careful: there is a real but modest signal, I do not yet have a replication, and I cannot say which model the brain uses. I think the most valuable parts of this work are the careful checks and the honest reporting of where each result stops.

Appendix A: Key Numbers

Item	Value
People / trials (ds004295)	23 people, 560 trials each, 12,880 trials
Behaviour (ds004295)	Win rate 0.604, stay rate 0.800, M1 alpha 0.405, beta 3.635 (means)
Blocks above chance (of 46)	M1 35, M2 39, M3 35
In-sample AIC winners (46 blocks)	M2 27, M1 15, M3 4
Out-of-sample winners (23 people)	M2 13, M1 6, M3 4; pairwise p = 0.30, 0.29, 0.83
RPE correlation between models	M1-M3 0.970, M1-M2 0.938, M2-M3 0.906
Raw decoding r (M1, CNN)	Frontal 0.080, central 0.066, parieto-occipital 0.047
Partial r after removing reward (M1)	-0.010, +0.019, -0.008 (all not significant)
PO interaction beta, seed 0 (raw)	M1 -0.4287 (p = 0.0025), M2 -0.0251 (p = 0.61), M3 -0.4881 (p = 0.0027)
Seeds significant (CNN)	M1 4 of 4, M2 0 of 4, M3 3 of 4
z-scored beta, mean of 4 seeds	CNN: -0.0180, -0.0022, -0.0205. Transformer 30 ep: +0.0010, -0.0025, -0.0014. Transformer 80 ep + early stop: -0.0218, -0.0066, -0.0129 (M1, M2, M3)
Jackknife	23 of 23 for M1 and for M3
ds003458 interaction	M1 beta -0.0168 (p = 0.1735); M3 beta -0.0166 (p = 0.1745); jackknife 0 of 23
Gradient boosting raw / partial r	HGB 0.136 / 0.023; AdaBoost 0.106 / 0.022
Gaussian mixture	3 groups (20, 2, 1), adjusted Rand index 0.27

Appendix B: Settings Used

Item	Setting
ds004295 EEG	BioSemi ActiveTwo, 1024 Hz, 66 channels raw; epochs -0.2 to 0.8 s (1025 points), baseline -0.2 to 0 s; regions 8 + 8 + 9 channels
ds003458 EEG	64 channels, 0.5 Hz high-pass, epochs -0.2 to 0.8 s (500 points at 500 Hz), baseline -0.2 to 0 s; regions 16 + 20 + 17 channels averaged to one trace each
RL model fitting	Several random starts, bounded optimisation. M1 alpha 0.001 to 0.999, beta 0.01 to 20
CNN (ds004295)	Conv1d 16 filters (kernel 15), Conv1d 32 filters (kernel 9), global average, linear; MSE loss; Adam 0.001; batch 32; 30 epochs; leave-one-person-out; seeds 0 to 3
Transformer	Conv stem (kernel 25, stride 5, 32 dimensions), learned position code, 2 layers, 4 heads, average, linear; AdamW; 30 fixed epochs, or up to 80 with cosine schedule and early stopping (patience 10) using one validation person

Item	Setting
CNN (ds003458)	3 conv blocks (16, 32, 32 filters) with batch norm, one input trace per region; MSE; Adam 0.001; 60 full-batch epochs; within-person 5-fold; seeds 0 to 3
Statistics	One-sample t-tests across people, paired tests, Wilcoxon, permutation test, Benjamini-Hochberg FDR (90-test grid), jackknife

Appendix C: Notebooks

Notebook	What it does
1. Data understanding and behaviour (ds004295)	Downloads and inspects the data, extracts events and epochs, first RL fits, the 'fits at chance' problem, tests of different choice codings, finding the authors' ground-truth files, matching them to people.
2. Dataset 2 (ds003458)	Reads the data from the public bucket, rebuilds trials from triggers, fits M1, M2, M3, parameter recovery, AIC and odd/even comparison, epoching with the 0.5 Hz filter, RPE traces, within-person CNN decoding, interaction test and jackknife.
3. Main analysis pipeline	Loads stored region tensors and fits, runs and reuses leave-one-person-out CNN and Transformer decoding, confound checks, seeds, jackknife, global FDR, exploratory analyses, out-of-sample comparison, and a status table.